

Ali Wael Mohamed Ali Ahmed

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Professional Summary

Passionate and curious Mechatronics Engineering Senior driven to take on new challenges, continuously learn, and contribute to purposeful, interesting projects. Strong foundation in **Robotics (ROS2, AI for Robotics)**, **Embedded C/C++ development**, **low-level driver implementation** (HAL, CAN, I2C, ADC, PWM), **Machine Learning**, and **Computer Vision**. Experienced in bridging physical constraints with efficient control logic across Robotics, Medical Devices, and electromechanical systems.

Education

Cairo University, Giza, Egypt B.Sc. in Mechatronics Engineering — Graduated: July 2026

Relevant Coursework: Control Systems, Automatic Control, Microprocessors & Microcontrollers, Thermodynamics, Embedded Systems, Advanced Robotics, Design of Mechatronics Systems, Electric Drive, Pattern Recognition & Artificial Neural Networks, Machine Intelligence.

Professional Experience

Embedded Linux Engineer Intern | Ezzmedical, Cairo, Egypt (Aug 2025 – Sep 2025)

Developed and deployed a highly reliable, custom Linux operating system for EZvent, the first Egyptian-developed medical ventilator to achieve unconditional regulatory approval.

Ensured system-level stability for life-critical operations, aligning with strict medical device compliance standards. Validated hardware-software integration by deploying a Qt-based interface onto ventilator boards, optimizing boot processes and low-level system performance.

Level 2 Automation Engineer Intern | Al Ezz Dekheila Steel Co. (EZDK), Alexandria, Egypt (Jul 2024)

Mapped complex industrial data flows between Level 1 and Level 2 automation hierarchies. Documented real-time data acquisition strategies to improve communication clarity across distributed industrial systems.

Technical Projects

Machine Learning Enhanced Emergency BVM Ventilator (Team Lead) (Sep 2025 – Present)

- **Embedded C/C++ & Architecture:** Designed Software Architecture Document & Software Requirements Specifications documents. Developed robust, safety-critical ventilation firmware in C/C++, managing real-time sensor calibration, 3 layer Architecture, serial communication, sensor filters and critical safety alarms.
- **Electrical Analysis, Integration & Wiring:** Designed & planned the full electrical system making sure of proper integration and compatibility between all components. Assembled all electrical components and accomplished a stable and safe wiring for the full electrical system.
- **Electromechanics & Motor Control:** Implemented advanced S-Curve motion profiling for stepper motors to ensure jerk-free operation, translating physical constraints and torque limits into efficient code.
- **Low-Level Drivers & Signals:** Designed sophisticated signal conditioning for pressure (MPX5010DP) and environmental (BMP280) sensors using ADC oversampling and Exponential Moving Average (EMA) filters to mitigate noise and drift.
- **Cross-Functional Agile Leadership:** Directed an interdisciplinary team (mechanical, electrical, software, medical), managing constraints of electromechanical integration.

Automated Bottling & Capping Line Simulation (Dec 2024 – Jan 2025)

- **PLC Programming & Automation:** Programmed Siemens PLCs in TIA Portal to simulate full bottling and capping operations with safety interlocks and reset logic.
- **HMI & Process Monitoring:** Designed a comprehensive HMI interface for real-time process monitoring and simulation of manual interventions, fully documenting industrial logic.

Robotic Arm Trajectory Automation & Teleoperation (Nov 2025 – Dec 2025)

- **CAN Bus Integration:** Configured low-level **CAN bus** network interfaces in Linux environments to monitor live telemetry and handle real-time connection states.
- **Control Laws:** Translated physical mechanics into software control laws, utilizing the robot's "teach mode" for high-precision trajectory tracking and HTTP requests for smartphone IMU teleoperation.

Minimalist Smartwatch on STM32 Microcontroller (Jun 2024)

- **Microcontroller Architecture:** Built a real-time digital watch programmed entirely in **bare-metal ARM Assembly** on an STM32F103C8T6.
- **Deep Hardware Understanding:** Demonstrated deep understanding of microcontroller architecture including memory mapping, direct register manipulation, interrupts, and hardware timers to achieve high accuracy. Interfaced a TFT display with optimized GPIO protocols.

Pick and Place Robotic Arm & Custom Gripper Control (April 2026 – May 2026)

- **Closed-Loop PID Control:** Engineered a custom robotic gripper mechanism driven by a lightweight (10g) open-loop DC motor. Implemented a closed-loop PID control system utilizing potentiometer feedback for precise position regulation and motor control.

- **Motion Profiling & Kinematics:** Developed synchronized multi-axis interpolation algorithms for the robot's servo motors, ensuring fast, smooth, and vibrationfree movement profiles.

Machine Fault Recognition using CNN (Mar 2026 – May 2026)

- **Deep Learning for Diagnostics:** Developed an end-to-end Convolutional Neural Network (CNN) model for machine fault recognition.
- **ML Pipeline Implementation:** Executed comprehensive data preprocessing, model training, and inference testing, demonstrating capabilities directly applicable to predictive maintenance and anomaly detection.

Autonomous Museum Guide Robot & AI Integration (Feb 2025 – May 2025)

- **Mechanical Design:** Designed the car chassis in SOLIDWORKS specifically for laser cutting. The design incorporated several flat parts to ensure we could utilize a single sheet of acrylic for cost efficiency and manufacturing speed. I designed the parts to have finger joints compatible with each other for easy assembly, and ensured proper holes were mapped out for the electronic components and the PCB.
- **Control & Sensor Fusion:** Integrated an MPU6050 IMU and encoders, applying PID control loops for stable, real-time autonomous navigation & Go To Goal logic, applying Aruco Markers Detection and Obstacle Avoidance using phone's camera and real time obstacle detection.
- **Machine Learning & Final Delivery:**
 - Deployed YOLOv11 object detection, enhancing real-time recognition latency by 30%.
 - Trained a custom YOLOv11 model on 3 specific museum artifacts using 50+ images per artifact covering different angles and lighting.
 - When the robot detects an artifact, it stops and utilizes a Text-to-Speech (TTS) integration to present the artifact to tourists.

Core Skills

- **Programming & Scripting:** C/C++, Python, ARM Assembly, MATLAB / Simulink.
- **Embedded Systems:** Embedded Linux, U-Boot, Kernel menuconfig, Microcontroller Architecture (STM32, ESP32, Arduino), Low-Level Software, Hardware Abstraction Layers (HAL), ADC, PWM, SPI, I2C, CAN Bus.
- **Control & Automation:** Control Theory (Automatic Control, PID, Kinematics), Motor Control & Motion Profiling, Electronics, Mechanics & Thermodynamics, PLC Programming (TIA Portal).
- **AI & Machine Learning:** Actively Exploring Agentic AI frameworks to automate C/C++ development and debug complex hardware/software issues. Practical experience building end-to-end ML pipelines (CNNs, Bayes Classifiers, OpenCV, YOLO) for fault recognition and predictive analytics.
- **Standards & Quality:** Familiarity with high code quality standards, strict compliance mindsets (Medical/Automotive), Agile methodologies.
- **Tools:** Keil, TIA Portal, Git, GitHub, SolidWorks, ROS, Linux, Matlab.

Certifications & Languages

- **Certifications:** Embedded Linux Course (IMT, 2023) | DELF B2 (French Ministry of Education, 2021)
- **Languages:** Arabic (Native), English (Proficient), French (Advanced/B2), Spanish (Beginner)